

Disrupt 141-C — Complete Project Archive

(v3.4)

- > **v3.4 update (September 2026).** Building map fixes only; the 3D model is unchanged.
- > - Six rooms (GF courtyard, board room, male toilet; FF workstation 16'-6" × 11'-8"; SF male bath, rooftop cafeteria) now colour their whole drawn outline, within ±10 % of the drawn area.
- > - Number markers never cover a label.
- > - Every size is complete, or reads "size not given on drawing".
- > - Details: the v3.4 part at the end of `HANDOFF\_v3.md`.

(v3.3 notes)

- > **v3.3 update (September 2026).** Room-by-room check against the key plans A-101 / A-102 / A-103 and the door / window schedules A-601 / A-602 / A-603.
- > - Every room is now labelled on the building map (zoom, pan, drawing underlay), in search (by name or size) and in the Architect view.
- > - Names and sizes follow the drawings; missing rooms and drawing notes (CB, 4S / 6S, exits, counters) were added.
- > - The first-floor "Stay lev." landing was added to the model, and the first-floor windows now sit on a 1'-0" upstand (A-602).
- > - Full table: `10\_v3\_rebuild/v3.3\_room\_check.md`. Details: the v3.3 part at the end of `HANDOFF\_v3.md`.

(v3.2 notes)

- > **v3.2 update (September 2026).** Adds the  **Architect view** for construction checks and  **Fly**.
- > - The Architect view shows a plan cut per floor, the registered drawings A-111/112/113 on each floor, wall hatch from the drawing legend, and existing and proposed columns.
- > - Room tags give the stated size and FFL. Hovering shows X/Y and height, and a tape measure reads to ½".
- > - Fly is a drone camera over and through the building.
- > - v3.1 before it corrected the coordinate frame and the north-west extension, traced the furniture from A-401/402/403, and added realistic people, department equipment and the generator.
- > - Nobody and nothing is placed outside the building except drawn open-air items.
- > **If you are an AI assistant picking this up: read the v3.1/v3.2 part at the end of `HANDOFF\_v3.md` first.**
- >
- > `04\_model/viewer\_standalone.html` is the v3.2 viewer.
- > `04\_model/Disrupt\_141C\_v3\_final.glb` is the final building model (cleaned walls, feet, Z-up) for Blender / Twinmotion / D5. The furniture and people are generated inside the viewer from `furniture.json`.

(v3 notes)

- > **v3 update (September 2026).** The model was re-checked against every plan sheet and rebuilt: the first and
- > second floors are now registered to the ground floor (they were ~4 ft out), the second-floor masjid is back,
- > the north-west compound is open ground, the new partitions are in, and every floor level follows the LEV tags.
- > **If you are an AI assistant picking this up: read `HANDOFF\_v3.md` first, then `HANDOFF\_v2.md`, then this file.**
- >
- > `04\_model/viewer\_standalone.html` is now the v3 viewer (v2 kept as `viewer\_standalone\_v2.html`).
- > `04\_model/Disrupt\_141C\_v3.glb` is the v3 model. `10\_v3\_rebuild/` holds the v3 pipeline and the verification overlays.

(v2 notes)

- > **v2 update (September 2026).** Everything from v1 is still here unchanged. v2 adds an interactive
- > walkthrough, a building directory map with wayfinding, furniture and people, lighting and a corrected model.
- > **If you are an AI assistant picking this up: read `HANDOFF\_v2.md` first, then this file.**

What's new in v2, in one place

`04\_model/viewer\_standalone.html`	**The v2 viewer.** Open in any browser, offline. (v1 kept as `viewer\_standalone\_v1\_original.html`.)
`04\_model/Disrupt\_141C\_v2\_corrected.glb`	Corrected model with all v2 geometry fixes baked in, for Blender / Twinmotion / D5. Same units (feet, Z-up) and object names; the two low-roof parapets are gone.
`08\_viewer\_v2/`	Viewer source, data and the walk/room/furniture pipeline. `build\_viewer.py` rebuilds the viewer exactly.
`09\_v2\_screens/`	Reference screenshots of v2.

v1 notes (original archive)

Everything from this project, self-contained. Nothing external is required.

If you are an AI assistant picking this up: read this file first, then
`03\_reports/FULL\_PROJECT\_REPORT.md`. That report contains the complete drawing

analysis and every modelling decision. You can continue the work from here without re-deriving anything.

What this project is

A 3D architectural reconstruction of the **Disrupt 141-C** building (Karachi) from a 23-sheet architectural working drawing set produced by Ingenious Design Studio, February 2026. The building is an existing three-storey courtyard office building being refurbished — the drawings are a survey plus proposed internal alterations, not a new-build design.

Status: complete. The model, renders and documentation are all finished. One open item remains (see "Open item" below).

Folder guide

Folder	Contents
`01\_source\_drawings/`	The original 23-page PDF working set — the source of truth
`02\_briefs/`	The two project briefs exactly as issued by the client, plus a PDF of them
`03\_reports/`	**FULL_PROJECT_REPORT.md** — drawing analysis + Parts A-G reconstruction record
`04\_model/`	The 3D model (GLB, OBJ) and a standalone offline viewer
`05\_renders/`	12 rendered views at 1920×1200
`06\_pipeline/`	Every script that produced the model — it is fully procedural and re-runnable
`07\_working\_data/`	Intermediate extraction data: wall masks, floor plates, calibration crops

The model

`04\_model/Disrupt\_141C.glb` — binary glTF, 26 named objects, 49,068 triangles.

`04\_model/Disrupt\_141C.obj` — universal fallback.

- **Units: feet. Up axis: Z.** Origin at the south-west corner of the compound.
- X runs east 0–151.08 ft, Y runs north 0–129.98 ft, Z is height above road datum.
- Opens in Blender, 3ds Max, Revit, Twinmotion, D5 Render, Unreal, or any web viewer.
- To scale to metres in Blender: multiply by 0.3048.

`04\_model/viewer\_standalone.html` — open this file in any browser, offline.

The model is embedded inside it. Orbit, toggle the seven layer groups, switch between eight camera presets, and switch between materials / white model / three floor cutaways / exploded axonometric.

Object names

GF_Slab	GF_External_Walls	GF_Glazing	GF_Internal_Walls	GF_Stairs
FF_Slab	FF_External_Walls	FF_Glazing	FF_Internal_Walls	FF_Stairs

SF_Slab	SF_External_Walls	SF_Glazing	SF_Internal_Walls	SF_Stairs
Roof_Level_12-5_Slab		Roof_Level_12-5_Parapet		
Roof_Level_23-10_Slab		Roof_Level_23-10_Parapet		
Roof_Level_35-3_Slab		Roof_Level_35-3_Parapet		
Roof_Corrugated_Sheet		Courtyard_Paving	Courtyard_Glazing	
Site_Boundary_Wall		Ground_Apron		

The key facts, in one place

Storeys	3, plus roof
Floor-to-floor	**11'-5"** on all three storeys (constant, stated)
Ground FFL	+1'-0" (west wing +2'-0")
First FFL	+12'-5" (west wing +13'-11" and +16'-11")
Second FFL	+23'-10" (west wing +25'-9")
Roof slab	+35'-3"
Parapet	+39'-3"
Clear height	10'-8" to soffit, 10'-2" under beams
Slab / beam	6" / 9"
Doors	8'-0" high throughout
Windows	7'-0" high on a 1'-0" sill
Stairs	5¾" riser, 1'-0" tread
Courtyard	23'-0" × 28'-4", full height, open to sky, glazed 34'-0"
Reception	20'-6" × 21'-5", 22'-1" high (double height)
Building envelope	≈ 148' × 127'-6" within a compound of ≈ 151' × 130'
Gross internal area	≈ 37,600 sq ft over three floors

How the geometry was derived (important)

Vertical geometry is exact. The level ladder on the elevations and sections is internally consistent to the inch and was used as drawn.

Horizontal geometry is calibrated, not dimensioned. The drawings are stamped NTS and contain **no overall perimeter dimension anywhere**. So the plan geometry was rasterised directly from the drawings rather than hand-modelled:

1. The dimension plans (A-111/112/113) colour-code existing walls in cyan and new walls in solid black — these layers were extracted by colour.
2. The reflected ceiling plans (A-501/502/503) fill the enclosed floor area in grey — this gives the floor plate per storey directly.
3. Both were converted at **41.70 pixels per foot at 500 dpi (1" = 12'-0")**, verified against three stated dimensions on three different sheets to better than 0.5%.
4. Facade openings were detected on the four elevations, each calibrated against its own level ladder (the four elevations are plotted at four different scales: 29.05, 31.72, 38.98 and 39.62 px/ft).

Independent cross-check that validates the whole approach: first-floor window heads and sills derived from the elevations landed at 13.36–20.42 ft, against 13.42–20.42 ft predicted from the schedules. A 0.06 ft agreement that was not built in.

Open item — the one thing that would improve this

There is no overall building dimension anywhere in the 23 sheets. The whole model's absolute scale rests on the calibration above.

One tape measurement of the front elevation length would convert this from a **calibrated** reconstruction to a **dimensioned** one. Because the pipeline is procedural, nothing else would need to change — edit the constant and re-run ``build_model.py``.

Also worth sending back to Ingenious Design Studio:

- Front parapet is tagged +38'-3" in the body of A-201 but +39'-3" in its own ladder and on A-202/A-203 (modelled as +39'-3")
 - The elevation names in the A-000 register are one sheet out of step with the sheet titles
 - Door schedules are numbered A-601/602/603 in the register but A-401/402(A)/403 on the sheets, colliding with the furniture plans
-

Rebuilding the model from scratch

The pipeline in ``06_pipeline/`` is complete and deterministic. Requirements:

```
python3 -m pip install numpy opencv-python pillow scipy trimesh shapely mapbox_earcut
playwright
# plus poppler-utils for pdftoppm
```

Order of operations:

```
pdftoppm -r 500 -png <source PDF> ... # render the sheets
python3 extract_plans.py # wall masks from A-111/112/113
python3 plates.py # floor plates from A-501/502/503, registered
python3 elevations.py # facade openings from A-201..A-204
python3 build_model.py # assembles and exports GLB + OBJ
python3 qa.py # QA data
python3 assemble.py # builds the standalone viewer
python3 shoot.py # renders the 12 views headlessly via Chromium
```

``build_model.py`` holds every decision as a named constant at the top of the file — levels, the courtyard and reception rectangles, stair positions, the sheet-roof zone. Change one and re-run; the whole model follows.

Known limitation: wall edges carry a faint 6" stepping because the plan is rasterised at 6" resolution. Reduce ``CELL`` in ``build_model.py`` for smoother silhouettes at the cost of triangle count.

Renders

File	View
01	Front elevation, near-orthographic — overlays A-201
02	Front three-quarter
03	Rear three-quarter
04	Left (west) side
05	Right (east) side — the service face
06	High three-quarter aerial
07	Eye level at the entrance
08	White-model axonometric
09	Ground floor cutaway (labelled)
10	First floor cutaway
11	Second floor cutaway
12	Exploded floor-by-floor axonometric

All rendered from the model itself. No image generation was used anywhere in this project.

Production rendering prompts

Ready-to-use prompts for Midjourney, D5 Render / Twinmotion, and Blender are in ``03_reports/FULL_PROJECT_REPORT.md``, Part F. They are written specifically for this building, with negative prompts for the features these tools habitually invent.

One safety note

The sections tag **asbestos cement sheet** as existing roofing at +35'-3". If this model is ever used to brief physical work, confirm the presence of asbestos with the consultant before anyone touches that roof.